

Interim Report Seda Kavlak - Protein Language Visualizer and RAG Agent

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Abstract

As climate change continues to impact marine biodiversity, understanding the biology of marine mammals at the molecular level is increasingly important. Proteomics can provide insights into population health, diversity, and ecological roles, but analyzing these complex datasets remains challenging. Dimensionality reduction techniques like UMAP can help to understand high-dimensional proteomic data, however most current tools are designed for expert users. This makes it difficult for non-specialists to explore and understand the information. To address this, I developed an interactive portal that visualizes proteomes of marine mammals in the Southern California region (“SoCal”) and integrates a Retrieval-Augmented Generation (RAG) AI agent. The portal combines UMAP-based visualizations with AI-generated explanations of taxonomic and protein-level data, making complex proteomic knowledge accessible to a broader audience. This approach demonstrates how visualization and RAG can work together to bridge the gap between expert-level data and public understanding, supporting conservation efforts and increasing engagement with marine science. - going to include results and discussion part later

Introduction

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Biodiversity loss

Human activity is causing a global loss of biodiversity [1]. The loss of species within an ecosystem can significantly alter its functioning [2]. Research shows that organisms play key roles in shaping their environments, for example through ecosystem engineering and by influencing the movement of elements in biogeochemical cycles (e.g., ecological stoichiometry). When biodiversity declines, these essential processes can be disrupted.

The marine realm is among the most affected by biodiversity loss [1]. Multiple factors are driving this decline. One major driver is climate change [3, 4]. Rising ocean temperatures lead to coral bleaching and create unsuitable conditions for many marine species [3]. In Australia, the giant kelp (*Macrocystis pyrifera*) is disappearing with ocean warming being one of the reasons [5]. In addition, increasing levels of atmospheric carbon dioxide are absorbed by the oceans, causing ocean acidification, which has harmful effects on marine organisms [1]. Climate change can also alter ocean currents, which are essential for distributing heat, regulating regional climates, and supporting

marine food webs.

Furthermore, the introduction of invasive species represents a serious threat to marine ecosystems [3, 6]. These non-native organisms are often transported through ship traffic, fisheries, canals, and other human activities, and they can outcompete native species for resources, leading to declines in local populations [3, 6].

Kimbrow et al. [7] demonstrated that regions dominated by native crabs and whelks have low native oyster mortality, whereas regions with high abundances of invasive species show much higher oyster mortality, resulting in less biologically diverse habitats. According to [6], virtually every coastal habitat in the San Francisco Bay area is dominated by at least one alien species. It should also be noted that many invasive species appear to benefit from rising temperatures due to climate change [8]. An example for this is the ascidian *Botrylloides violaceus* shows increased growth and more frequent reproductive cycles as temperatures rise. This temperature-driven boost in reproduction and growth gives it a competitive advantage in coastal communities.

Beyond their ecological impacts, invasive species also affect human societies by reducing resources such as fisheries, aquaculture, tourism, and marine infrastructure [6]. Consequently, these impacts can have serious social and economic effects, including losses in employment in sectors dependent on marine ecosystems.

Urbanization also negatively affects marine biodiversity [3]. It can result in increased water pollution, which harms marine species, and air pollution, which further contributes to climate change. In addition, the use of pesticides can degrade aquatic environments. A well-known example of these combined pressures is the degradation of Australia's Great Barrier Reef. Similarly, [9] reports an average loss of 26 percent in urban areas.

Finally, marine wildlife is heavily impacted by plastic pollution, with fatalities recorded in 80 species [3]. Macroplastic debris poses a major threat to marine life through entanglement and ingestion [10]. Records show tens of thousands of individuals across hundreds of species have been affected, with entanglement—mainly in fishing nets and ropes—being more common than ingestion. Both can cause serious injury or death, making macroplastics a significant cause of mortality in coastal and marine ecosystems. This highlights how human activities continue to place severe and increasing pressure on marine ecosystems. For these reasons conservation efforts gain increasing importance.

One of the conservation efforts currently taking place in California is the California Conservation Genomics Project [11]. This project generates comprehensive landscape-level genetic data for at-risk species, helping identify resilient populations and critical habitats to inform management and conservation planning. By sequencing and analyzing genetic variation across species and environments, this project aims to enhance long-term species survival in the face of climate change and other threats. Another project in California is the Marine Life Protection Act Initiative [12]. This initiative established a statewide network of marine protected areas designed using scientific guidelines to conserve marine biodiversity, rebuild depleted populations, and protect ecosystem structure and function. It represents a large-scale, ecosystem-based approach to marine conservation that integrates science, policy, and stakeholder involvement.

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Link between conservation efforts and education

The success of conservation initiatives depends not only on scientific and policy efforts, but also on public awareness and education. Multiple studies suggest that education and environmental knowledge are positively associated with pro-environmental attitudes and support for conservation initiatives [13]. For example, research in local communities in Indonesia has shown that people with higher levels of education exhibit more favorable attitudes toward biodiversity conservation [14]. Similarly, a study of Indian students found that interest in nature conservation increased after learning about their local environment [15]. Furthermore, [16] concluded that there is a direct and significant relationship between environmental education and pro-environmental behavior among basic education students in the Peruvian Amazon. This highlights the importance of education the general public on the environment and its conservation.

One promising approach to simultaneously support scientific research and foster public environmental education is citizen science. Citizen science is the practice of public participation in scientific research [17]. By partnering with professional scientists, members of the public can collect, submit, or analyze data. In addition to contributing to research, participants gain knowledge about scientific concepts and processes, while also increasing public awareness of environmental issues. For these reasons conservation initiatives that leverage citizen science to fill research gaps are becoming increasingly important [18, 19]. This approach has proven effective in marine conservation. For example, studies on fish spawning aggregations (FSAs) on Caribbean coral reefs show that sites monitored with the participation of local fishers are more likely to receive protection than those studied without community involvement [20]. One example of utilizing citizen science in California is Reef Check California [21]. Reef Check California is developing a network of engaged and knowledgeable citizens dedicated to supporting the sustainable use and conservation of California's nearshore marine environment. To achieve this, volunteers are trained as citizen scientists to monitor nearshore reefs and collect data on the status of key indicator species.

While citizen science and public education demonstrate the value of making scientific research accessible and engaging to the public, much of this work operates at the level of species observations, habitats, or ecological patterns. In contrast, modern biological research increasingly relies on large-scale molecular data, such as genomic and proteomic datasets [22]. These type of data are often referred to as "omics". The term "omics" describes large-scale, data-rich approaches in biology, encompassing fields such as proteomics, transcriptomics, genomics, metabolomics, lipidomics, and epigenomics [23]. As a rapidly growing and transdisciplinary domain, omics research enables the integration of information across multiple molecular layers, thereby providing multi-level insights into biological systems that were previously not possible. The combination of molecular data with large-scale environmental information, such as remote sensing, can further transform our understanding of biodiversity patterns and improved predictions of how ecosystems respond to climate change [24]. The previously mentioned California Conservation Genomics Project, for example, highlights the importance of genomics in conservation by using genetic data to identify resilient populations and inform management strategies [11]. The use of genetic information in this way demonstrates how modern omics-based approaches can support more effective and evidence-based biodiversity stewardship. Furthermore, proteomics play a significant role in marine life conservation. The proteome is the entire collection of proteins expressed by a biological system at a certain time [25]. Because the proteome is dynamic, reflecting changes over

time and in response to environmental events, it can provide critical insights into organismal health and ecological interactions. In marine mammals, for example, proteomes can provide information on their diversity, population health, and ecological roles, which in turn can inform conservation strategies [26–28].

Building on this, proteomics research has relevance beyond conservation. Some examples include identifying medically important human pathogens, analysis of animal venom, gaining a deeper understanding of eukaryotic genome organization and regulation, inferring plant resistance to pests and diseases, and identifying co-evolution of RNA and protein levels [29–33].

Despite this growing importance and impact, the complexity and scale of omics data pose significant challenges. In a comprehensive literature review, [34] examined public knowledge of omics data and reported that most study participants demonstrated only a very limited understanding of these types of data. Similarly, a study on the practical application of omics data found that even specialists, such as dentists, are not being adequately prepared during their education to interpret and apply omics-based findings [23]. Hence, this type of research remains largely inaccessible to non-specialist audiences [34].

Taxonomy and Phylogenetics

Taxonomic understanding is critical for studying biodiversity [35]. Taxonomy is the biological discipline concerned with identifying, describing, classifying, and naming species and other taxa [36], and thus defines the fundamental units used to organize, evaluate, and manage biodiversity in both research and conservation [35]. Its importance is evident in the management of invasive species, where inconsistent taxonomic treatments can delay detection and control, as seen in the cactus genus *Opuntia*. Similarly, species assessments are complicated by taxonomic inconsistency, since many species exist under multiple names, making it difficult to integrate ecological and distributional data and to reliably track threats and population declines.

Beyond classification, understanding evolutionary relationships between species is also crucial for conservation. Phylogenetics is the science of reconstructing the evolutionary history of life on Earth [37]. It uses a combination of morphological and genomic data to construct phylogenetic trees that display relationships between taxa. These species phylogenies can be used, for example, to assess whether communities are phylogenetically clustered or dispersed relative to random expectations [38]. Phylogenetically dispersed communities tend to contain greater evolutionary diversity and may therefore be more resilient to stressors such as pests, pathogens, and climate change, whereas phylogenetically clustered communities are often more vulnerable to such disturbances. Moreover, evolutionary relationships can inform decisions about human intervention in ecosystems; for example, [39] discuss how human-induced hybridization between closely related species may, in some cases, enhance adaptability to changing environmental conditions.

Phylogenetic relationships can also be observed on proteomics. Proteins are composed of domains, which can be modified, duplicated, or transferred during evolution, making them useful units for studying protein structure, function, and evolution [40]. With the growth of genome sequencing, proteome-based analyses using protein domain content have become widespread. Phylogenies inferred from genome-wide domain content are often comparable to those derived from gene sequence data, and such analyses can also reveal the evolutionary history of domains and identify taxa-specific domains. [40]

showed that protein domain clusters can be identified based on similarities in their usage across species and that these clusters are associated with distinct taxonomic groups as well as characteristic structural and functional properties. More recently, large-scale clustering of AlphaFold protein structures has grouped millions of proteins into evolutionarily meaningful clusters [41]. Many of these clusters lack functional annotation and likely represent previously undescribed folds. These analyses show that most clusters are ancient in origin. A smaller fraction appears to be species-specific. This further highlights how protein clustering can reveal both deep evolutionary relationships and lineage-specific novelty.

Together, these studies show that evolutionary and taxonomic relationships are reflected in protein sequences, domains, and structures, and that clustering approaches can reveal both deep evolutionary patterns and lineage-specific differences. However, exploring such relationships at the scale of entire proteomes remains challenging due to the size and high dimensionality of protein datasets [42].

Transformer Models and Protein Language Models

To address the challenge of representing complex protein data, protein sequences can be transformed into representations that capture their underlying biological properties. This can be achieved by leveraging Protein Language Models (PLMs) [43]. PLMs are inspired by Natural Language Processing (NLP) models and use transformer architectures to process protein sequences in a manner analogous to how NLP models process human language, treating amino acid sequences similarly to sentences [44, 45]. PLMs are typically pretrained on large protein sequence datasets [44, 45]. By using transformer architectures with self-attention mechanisms, these models can learn the “grammar” of protein sequences, including structural motifs, evolutionary relationships, and functional patterns. Self-attention mechanisms generate representations of individual sequence elements by incorporating information from all other elements within the same sequence [46].

The transformer architecture can be described as follows. First, the input sequence is tokenized, meaning it is broken down into smaller units [47]. Each token is then mapped to an embedding vector, which is a numerical representation of the token. Because the meaning of a token depends on its context within the sequence, self-attention is used to allow the model to consider relationships between all tokens. In this way, the model can determine which parts of the sequence are most relevant when updating the representation of each token. As a result, embeddings with similar meanings tend to be located close to each other in the vector space. Additionally, the dimensionality of an embedding vector reflects the number of features it can encode. For example, a vector of length three could represent properties such as evolutionary relatedness, structural similarity, and functional similarity [48].

After pretraining, the learned self-attention layers can be applied to previously unseen protein sequences. When a sequence is passed through the model, it is transformed into a set of high-dimensional embeddings that encode contextual, structural, and evolutionary information derived from the entire sequence. These embeddings provide a compact numerical representation of protein sequences and can be used as input for downstream analyses. Although transformer models often include a decoder component for sequence generation tasks, in this work we will only focus on the encoder part of the model as it is used to create vector embeddings. PLMs have been successfully applied to a variety of tasks, including protein structure prediction, sequence modeling, and functional protein generation across diverse protein families [49–51].

Dimensionality Reduction Methods

Visualizing these high dimensional embeddings obtained by PLMs requires dimensionality reduction techniques [52]. One common approach to dimensionality reduction is Principle Component Analysis (PCA) [53]. After scaling and log transformation, the data are linearly transformed into a new coordinate system using principal component analysis (PCA), such that the first component captures the largest fraction of the overall variability and subsequent components progressively less. This aggregates most of the data variance into fewer dimensions, enabling visualization of high-dimensional data in two dimensions. In addition, the contribution of the original variables to each principal component can be examined, and the variance explained by each component provides information about the relationships among the underlying variables, with a single dominant component indicating partial dependence and a more even distribution suggesting greater independence. However, this approach does not always produce a clear visual separation between proteins, particularly when the underlying structure of the data is highly nonlinear.

When dealing with non-linear relationships, t-Distributed Stochastic Neighbor Embedding (t-SNE) provides a solution [54]. t-SNE computes low-dimensional embeddings of high-dimensional data points. First, Euclidean distances between points are converted into probabilities representing their pairwise similarity [55]. The method then seeks a low-dimensional representation in which points with high similarity remain close together, while dissimilar points are placed farther apart, thereby preserving local neighborhood structure and revealing cluster-like patterns in the data. This is achieved by minimizing a loss function that measures the difference between these probability distributions.

The widescale use of t-SNE inspired the development of other non-linear dimensionality reduction algorithms [56]. UMAP operates in two main phases [57]. In the first phase, graph construction, the algorithm identifies the k nearest neighbors for each observation and applies local distance normalization to compute similarity weights, thereby capturing the local structure of the high-dimensional data. In the second phase, graph layout optimization, UMAP initializes a low-dimensional embedding and then iteratively updates it using gradient descent by pulling connected points closer together while pushing randomly sampled non-neighbor points apart to preserve the graph's structure in the low-dimensional space. While UMAP and t-SNE deliver similar results, UMAP preserves more of the global structure with better run time performance [58]. Furthermore, the algorithm is able to scale to significantly larger data set sizes than are feasible for t-SNE. Lastly, UMAP has no computational restrictions on embedding dimension.

UMAP has been successfully applied across multiple biological domains. For example, [59] utilized UMAP to analyze proteomics data from colon adenomas, while [60] employed it to visualize relationships between organs in mouse and rat tissues.

RAG as an education tool

Despite the increasing availability of proteomics analysis and visualization software, most current tools remain oriented toward expert users with strong computational or domain-specific training, limiting accessibility to non-specialists [61]. Additionally, as previously mentioned, scientific information is often difficult for non-specialists to understand, which can reduce public engagement and participation in conservation efforts [62]. Large Language Model (LLM) powered chatbots like ChatGPT, DeepSeek

or Meta are already being widely used by experts and the general public [63,64]. However, these LLMs are known to hallucinate [65]. When hallucinating the model may generate irrelevant or random outputs that may hold no correlation with inputs or desired outputs. To solve this problem and increase accuracy, Retrieval Augmented Generation (RAG) can be used [66]. RAG can help increase the accuracy of a Large Language model (LLM) by acting as non-parametric memory for the LLM, thus providing external knowledge that can be updated and expanded [67,68].

There are different RAG approaches, but the simplest approach consists of three main processes: indexing, retrieval and generation [69]. First, during raw indexing, the data gets cleaned and extracted into plain text. The text is segmented into chunks that are then encoded in a vector representation. The vectors are then stored in an indexed vector database for an efficient similarity search in the next phase. The retrieval phase first starts with a user query. The model encodes the query into a vector representation similar to the indexing step. Then it computes similarity scores between the query vector and the chunk vectors. The top k chunks with the greatest similarity to the query are retrieved. Finally, in the generation stage, the retrieved chunks are combined with the original query (often by augmenting the prompt) and fed into a large language model. The model uses this context plus its own learned knowledge to generate a response grounded in the retrieved information.

[70] built a RAG driven education tool called HiTA, which is an AI-assisted learning system that puts educators in control by using LLMs as teaching assistants under pedagogical supervision, delivering course-specific and instruction-aligned responses. Another RAG-powered tool is OwlMentor, an AI-powered learning environment designed to assist university students in comprehending scientific texts [71]. Specifically, in the biological space [72] leveraged this approach to create automated summaries of biomedical research that is easy to understand for the general public. Furthermore, [] built BioRAG which is a life-science-focused RAG framework that indexes 22 million papers, uses domain-specific embeddings, hierarchical and iterative retrieval, and outperforms existing LLM and RAG approaches on multiple life science question-answering tasks. However, there has not been a RAG that specifically explains taxonomic, phylogenetic and proteomic data.

In this work, I present the Protein Language Visualizer (PLVis), an interactive, conservation-oriented platform that tightly integrates large-scale proteome visualization with a Retrieval-Augmented Generation (RAG)-powered assistant. PLVis allows users to explore and compare marine mammal proteomes from the Southern California region through intuitive visual representations, such as overlapping proteomes, domain structures, and reduced-dimensional embeddings, while simultaneously querying an AI agent about what they are seeing. Rather than treating visualization and explanation as separate steps, my system couples them: users can select species (e.g., dolphin and orca), inspect shared and distinct protein features, and ask context-specific questions such as what particular proteins or domains represent and what biological roles they play. The RAG agent grounds its responses in curated scientific sources, reducing hallucinations and providing explanations that are both accurate and accessible to non-specialists. In doing so, PLVis bridges the gap between complex proteomic data and public understanding, offering a unified tool that supports education, exploration, and conservation-relevant insight into biodiversity at the molecular level.

Materials and methods

U-Map Visualization

I downloaded the proteomes of five marine mammal species found in Southern California: *Eschrichtius robustus* (gray whale), *Orcinus orca* (killer whale), *Zalophus californianus* (California sea lion), *Phoca vitulina* (harbor seal), and *Tursiops truncatus* (bottlenose dolphin). Proteome datasets were obtained from either NCBI or UniProt; full source details are provided in Table 1.

Table 1. Marine mammal genome and proteome data sources. Genomic and proteomic resources used in this study, including organism names, data type availability, source database, last modification date, and download links.

Common name	Scientific name	Data type	Source	Last modified	Link
Gray whale	<i>Eschrichtius robustus</i>	Proteome	NCBI	Sep 2024	NCBI FTP
Orca	<i>Orcinus orca</i>	Proteome	NCBI	Mar 2023	NCBI FTP
Californian sea lion	<i>Zalophus californianus</i>	Proteome	UniProt	May 2025	UniProt Proteome
Harbor seal	<i>Phoca vitulina</i>	Proteome	NCBI	Nov 2020	NCBI FTP
Bottlenose dolphin	<i>Tursiops truncatus</i>	Proteome	UniProt	May 2025	UniProt Proteome

I selected these species for the initial version of our proteome visualiser for local flora and fauna because marine mammals are highly visible in Southern California and play a significant role in attracting tourism [73]. Species such as the harbor seal and cetaceans like the bottlenose dolphin, killer whale and gray whale are considered umbrella species; protecting them indirectly protects other species that rely on them as predators, prey, or ecological indicators [74, 75]. Additionally, reference proteomes were available for all selected species.

Each FASTA file, representing the reference proteome of a specific organism, was annotated using SeqHub (formerly known as Gaia) [76]. An example of the resulting CSV table can be seen in Table 2. Each row represents a single protein and its annotations. The **Entry** column provides the unique protein identifier from the UniProt database, while **Sequence** contains a preview of the first 20 amino acids of the full sequence. The **Protein Name** column gives a descriptive name of the protein, derived from the SeqHub **desc** field. Functional domains identified via Hidden Markov Models (HMMs) are listed in **Domains (hmm.labels)**, with corresponding Pfam identifiers in **Pfam IDs (hmm.pfam.ids)**, their positions along the sequence in **Domain Ranges (hmm.ranges)**, and plain-language descriptions in **HMM Descriptions (hmm.descriptions)**. The **Cluster** column contains the K-means cluster label assigned during UMAP construction, indicating which group of functionally similar proteins the sequence belongs to.

Table 2. Example rows from the annotated Bottlenose Dolphin proteome as displayed in the protein selection table. Sequence is truncated to the first 20 amino acids; Protein Name is trimmed to remove the OS= suffix.

Entry	Organism	Protein Name	Sequence	Cluster	Domains	HMM Descriptions	Pfam IDs	Domain Ranges
A0A2U3UYK0	Bottlenose Dolphin	Peroxisomal coenzyme A diphosphatase NUDT7	MWRSSPPQKPVR	NSLIDD..	NUDIX	NUDIX domain	PF00293	39–182
A0A2U3UYS0	Bottlenose Dolphin	Norrin	MRNHVLAASFMSLLAL...	DAN;	Cys_knot	DAN domain; Cystine-knot domain	PF03045; PF00007	18–131; 37–133
A0A2U3UZ09	Bottlenose Dolphin	Calreticulin	MLLPVLVLLGLLGLGAAE...	Calreticulin		Calreticulin family	PF00262	23–258; 255–332
A0A2U3UZ24	Bottlenose Dolphin	Mitochondrial ubiquitin ligase activator of NFKB 1	MESGGRPSLGQFHLLGTS...	GIDE; zf-C3HC4_3		E3 Ubiquitin ligase; Zinc finger C3HC4	PF12483; PF13920	95–256; 298–346
A0A2U3UZF7	Bottlenose Dolphin	B-cell lymphoma/leukemia 10	MEPTAPSLTEEDL	TEVKK...	CARD	Caspase recruitment domain	PF00619	18–102

Additionally, all protein sequences were embedded using ProtT5-XL [77]. ProtT5-XL is a transformer-based protein language model pre-trained on UniRef50, a large clustered database of protein sequences. Rather than comparing sequences character by character, it generates a fixed-length embedding vector for each protein that captures structural, functional, and evolutionary properties implicitly. This means two proteins can have similar embeddings and therefore appear close together in the UMAP. These embeddings were then used to create UMAP visualizations.

Because UMAP projections are only comparable when computed in the same embedding space, I generated a single shared UMAP by combining the protein embeddings from all species. After computing the UMAP coordinates and performing k-means clustering on the embeddings, I extracted and saved species-specific subsets while preserving their coordinates in the shared space. This approach ensures that UMAP visualizations remain directly comparable across species, allowing the proteome explorer to display individual species or any combination thereof while maintaining meaningful spatial relationships.

RAG-Agent

My goal was to create an RAG agent that would serve as a well-rounded assistant for PLVis users. I anticipated that users would ask questions across three main areas: the UMAP visualization and how to read and interpret it, the taxonomic relationships among the featured species, and questions related to specific proteins or protein clusters. For this reason, the RAG agent was provided with academic papers and books about marine mammals and proteomes. A list of literature can be found in Table A in the appendix.

To allow efficient information retrieval, I first segmented the literature into discrete units called “chunks” based on document structure. For HTML documents, each chunk corresponds to a section defined by a heading (e.g., “III Ecology” or “V Life History”). When a new heading is encountered, the current chunk is terminated and a new one is initiated. Additionally, a 4,000-character safety limit is enforced; if a section exceeds this threshold, it is further subdivided to ensure compliance with the limit. This additional subdivision is necessary because excessively large chunks tend to contain substantial amounts of irrelevant or peripheral information. Such noise can negatively affect retrieval performance, as it becomes more difficult for the large language model (LLM) to identify the most semantically relevant chunk in response to a given query. By limiting chunk size, the retrieval process becomes more precise and contextually focused. For PDF documents, the same 4,000-character constraint is applied. As PDFs do not contain explicit structural tags comparable to HTML headings, section boundaries are identified heuristically. Specifically, any text span rendered in bold typeface on a given page is interpreted as a section heading and treated as the start of a new chunk. To further reduce noise in the corpus, I applied filtering rules during extraction. Repeated short text blocks (appearing more than three times and under 200 characters) were excluded as running headers or footers. Figure and table captions were omitted. Text processing was halted upon encountering terminal sections such as References, Bibliography, Acknowledgements, Funding, or Supplementary Material, as these do not contribute meaningful domain knowledge for retrieval. Images embedded within the documents were extracted and processed separately. Tables were extracted as TSV. Since images cannot be embedded directly, each image was described using the gpt-4o mini model, and the resulting textual summaries were processed as a chunk each. Finally, 2,628 chunks total (2,499 text, 32 tables, 97 image summaries) were generated. Each chunk includes structured metadata (document name, chapter, section title, and

page number). This metadata supports more precise and context-aware retrieval.

The extracted chunks were subsequently transformed into dense vector embeddings using OpenAI's text-embedding-3-small model. Each resulting embedding consists of a 1,536-dimensional vector representation, where each dimension captures latent semantic features of the respective text segment. As a result, chunks discussing related biological concepts are positioned in close proximity within the shared high-dimensional vector space. The generated embeddings, together with their corresponding raw text and structured metadata were saved using ChromaDB [78]. ChromaDB is an open-source vector database designed to store and efficiently retrieve high-dimensional embeddings. At query time, the user's question undergoes the same embedding procedure using the identical model, producing a query vector within the same 1,536-dimensional embedding space. ChromaDB then performs a cosine similarity search over all indexed vectors and retrieves the $k = 8$ most semantically relevant chunks. Cosine similarity evaluates the angular distance between vectors rather than their Euclidean magnitude, making it less sensitive to differences in text length and vector norm. Importantly, employing the same embedding model during both indexing and retrieval ensures that the learned geometric relationships within the vector space remain consistent. Using different embedding models at either stage would result in incompatible vector representations, thereby invalidating similarity comparisons. The retrieved chunks, each including metadata, are subsequently provided as contextual input to the language model, allowing the generation of responses. The generation prompt additionally instructs the model that if the retrieved chunks are not sufficiently relevant to the query, it should draw on its general marine biology knowledge while explicitly flagging the response as not sourced from the indexed literature. This ensures the system degrades gracefully rather than returning empty or misleading answers. The resulting raw answer is then passed through a second language model call, which reformulates the response into an accessible, conversational register appropriate for the target audience and appends contextually relevant follow-up suggestions.

To make conversations with the RAG agent seem more natural, I implemented a "router" LLM. When a user submits a query, it is first passed to this router, which classifies the incoming message as either conversational (i.e. "Hi", "Cool!", "Wow" etc.), visual (related to what the user can see on screen or the UMAP) or a biology question. If the question is conversational the RAG call is not made and gpt-4o mini is used to reply. If the query is biology related a RAG call is made and the previously described pipeline is executed. However, if the query is visual, no retrieval is performed. Instead, a pre-assembled text description of the current screen state is passed directly to the LLM as context. This visual context is assembled on every message and has three components. The first is a spatial overview, which is always present. The UMAP plot is divided into a 3×3 grid and the number of proteins from each species landing in each region is counted, along with any regions where species overlap. The LLM receives this as a readable summary rather than raw coordinates. The second component is a cluster summary. The top 30 clusters are listed in descending order of size, each labelled as either unique to a single species or shared across multiple, and accompanied by a small sample of representative protein names. Together, these two components give the LLM a compact but informative picture of the current state of the plot at all times. The third component is only present when the user has lassoed a selection. Rather than sampling a fixed number of individual proteins, a single pass over all selected points is performed. Then three things are computed: the top 10 most frequent Pfam/HMM domains across the entire selection (providing the most informative signal for questions like "what do these proteins have in common?"), a breakdown of the top 10 clusters by size each

labelled as unique or shared across species, and 5 representative sample entries with their UniProt ID, protein description, UMAP coordinates, and cluster assignment. This design keeps the prompt size roughly constant regardless of selection size and a lasso of 500 proteins, for example, produces a comparably sized context string as a lasso of 20.

RAG Evaluation

To assess whether my RAG agent outperforms its base model, gpt-4o mini, in answering biological questions, I conducted an evaluation following [66]. For this, I decomposed the RAG system into its retrieval and generation components.

The retrieval module is responsible for fetching relevant information that informs the subsequent generation process [66]. A key challenge in evaluating retrieval is measuring how well the retrieved documents align with the query in terms of relevance and coverage. This difficulty is compounded by the dynamic nature of knowledge sources, where information may change over time, introducing temporal uncertainty. Moreover, the heterogeneity and varying quality of sources increase the likelihood of retrieving incomplete or misleading content. Consequently, traditional metrics such as precision and recall alone are insufficient to fully capture retrieval performance in RAG systems, necessitating more nuanced and task-specific evaluation approaches.

The generation component, produces responses based on the retrieved content. Its evaluation focuses on faithfulness to the retrieved information, factual accuracy, relevance to the query, and coherence of the output [66]. This assessment becomes more challenging for open-ended questions, where defining a single “correct” answer is inherently subjective.

Furthermore, the RAG system must be evaluated as a full system due to the interplay between retrieval and generation [66]. In particular, the contribution of the retrieved content to the final response should be assessed. Additionally, factors such as response latency and the ability to handle ambiguous or complex queries must also be considered.

As [79] has noted, blindly following prior evaluation practices can introduce uncertainty about why a given metric was chosen and what it was intended to measure. I therefore adopted a minimal approach, selecting only metrics I could rationalize for my specific use case.

Before running the evaluation, I created a ground truth dataset using AutoRAG, an open-source framework designed to automate the construction and evaluation of RAG pipelines [80]. Two hundred questions were generated by randomly sampling single chunks from the corpus and prompting GPT-4o-mini to produce factoid questions grounded in each chunk’s content. For each question, GPT-4o-mini then generated both a detailed and a concise ground-truth answer. Questions for which the model could not produce a confident answer were removed using a rule-based filter, ensuring the final dataset contained only well-formed, answerable questions. Each QA pair also records the chunk ID from which the question was derived.

The evaluation parameters and their definitions can be found in Table 3. I evaluated retrieval performance three metrics. Recall measures the proportion of relevant instances successfully retrieved out of all relevant cases [66]. Two variants of this were used: Recall@1, which evaluates whether the correct answer appears as the top-ranked result, and Recall@8, which considers whether it appears anywhere within the top eight retrieved chunks. Finally, Mean Reciprocal Rank (MRR) captures the average rank position of the correct answer across queries, rewarding systems that surface relevant

results higher in the ranking. To assess generation quality I chose two metrics. BERTScore evaluated how semantically aligned the generated answer was with the ground truth by computing contextual embeddings via BERT and measuring their token-level similarity against the ground-truth answer [66]. SemScore assessed sentence-level similarity by computing the cosine similarity between the generated response and the ground-truth answer [81]. Together, BERTScore and SemScore provide complementary perspectives on generation quality: BERTScore captures fine-grained token-level lexical overlap, while SemScore reflects the overall semantic similarity of the response as a whole. For both metrics, I additionally report a win rate, defined as the proportion of questions for which the RAG system achieved a higher score than the base model. To ensure the RAG agent stayed true to the retrieved chunks when generating an answer, faithfulness was evaluated using an LLM (GPT-4o-mini) as judge [66].

Table 3. Evaluation metrics for retrieval and generation.

Metric	Type	Formula / Definition
Recall@1	Retrieval	$\text{Recall@1} = \frac{ Rel \cap Ret@1 }{ Rel }$
Recall@8	Retrieval	$\text{Recall@8} = \frac{ Rel \cap Ret@8 }{ Rel }$
MRR	Retrieval	$\text{MRR} = \frac{1}{ Q } \sum_{q \in Q} \frac{1}{\text{rank}_q}$
BERTScore	Generation	Cosine similarity between contextual token embeddings of a pre-trained BERT model
SemScore	Generation	$\text{sem_score} = \frac{\mathbf{a} \cdot \bar{\mathbf{r}}}{\ \mathbf{a}\ \ \bar{\mathbf{r}}\ }, \quad \bar{\mathbf{r}} = \frac{1}{N} \sum_{i=1}^N \mathbf{r}_i$
Faithfulness	Generation	LLM-as-Judge evaluation using GPT-4o-mini

Additionally, I evaluated latency separately for the RAG pipeline (embedding, Chroma retrieval, GPT generation) and the base model (GPT generation only) and then averaged across all 200 questions to report mean latency and RAG overhead.

Results

UMAP

Examples of the resulting UMAP plot can be seen in Fig. 1 and Fig. 2. Figure 1 shows the UMAP projection of the proteomes of all five species: California Sea Lion (*Zalophus californianus*), Bottlenose Dolphin (*Tursiops truncatus*), Gray Whale (*Eschrichtius robustus*), Killer Whale (*Orcinus orca*), and Harbor Seal (*Phoca vitulina*). The data points from all five species are distributed across a single, continuous cluster with no clear separation between species. Points of all five colours are present throughout the full extent of the cluster, including both the dense central region and the sparser periphery. No species is visibly isolated from the others in the embedding space.

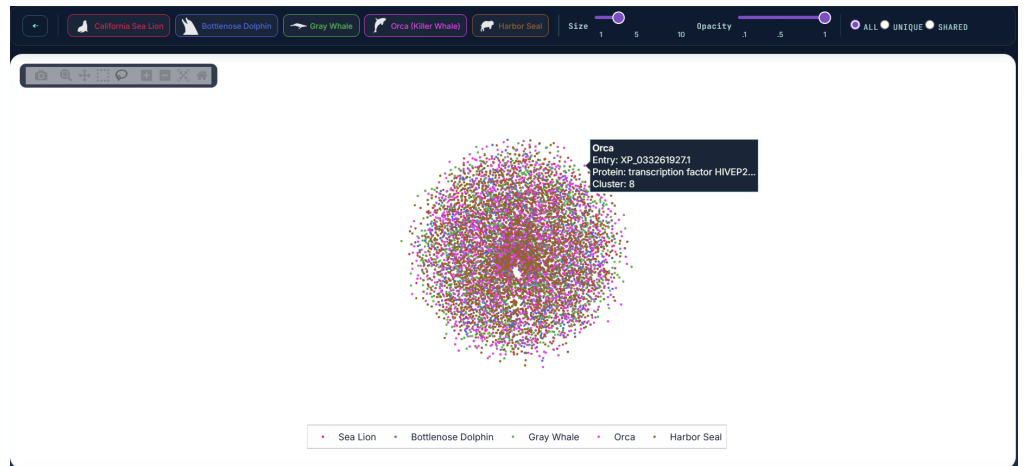


Fig 1. UMAP projection of the proteomes of all five species (Harbor Seal, Bottlenose Dolphin, California Sealion, Gray Whale, Killer Whale)

Figure 2 shows the UMAP projection of the proteomes of Bottlenose Dolphin (*Tursiops truncatus*) and Killer Whale (*Orcinus orca*). The two species display substantial overlap across the embedding space. A dense central region is visible where the two populations co-localise, with points from both species distributed throughout. A small number of data points are scattered toward the periphery.

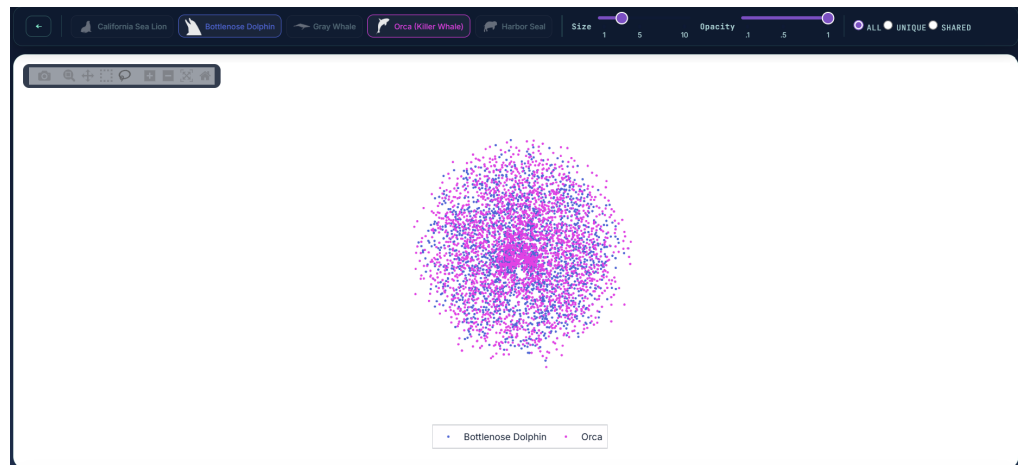


Fig 2. UMAP projection of the proteomes of all five species.

The number of proteins per species is shown in Table 4. The Killer Whale has the largest proteome in this dataset, while the Gray Whale has the smallest. Differences in proteome size reflect variation in genome assembly completeness and annotation depth rather than true biological differences in gene count. Across all five species, the combined dataset comprises 443,129 proteins.

Table 4. Number of proteins per species included in the UMAP visualisation.

Species	Proteins
California Sea Lion (<i>Zalophus californianus</i>)	86,604
Bottlenose Dolphin (<i>Tursiops truncatus</i>)	82,376
Gray Whale (<i>Eschrichtius robustus</i>)	78,421
Killer Whale (<i>Orcinus orca</i>)	112,335
Harbor Seal (<i>Phoca vitulina</i>)	83,393
Total	443,129

RAG Evaluation

The results of the RAG evaluation can be seen in Table 5. Regarding retrieval, Recall@8 reached 0.855, meaning the correct chunk was present in the top eight retrieved results for 85.5% of queries. Recall@1 was lower at 0.42, and MRR was 0.60, indicating that while the correct chunk was frequently included in the candidate set, it was not always ranked first. In the generation evaluation, the RAG model outperformed the base model by 0.10 in SemScore (0.818 vs. 0.714) and by 0.031 in BERTScore (0.926 vs. 0.895), indicating that RAG-augmented responses were more semantically aligned with the ground truth answers than those produced by the base model alone. The RAG system outperformed the base model on 81.5% of questions by SemScore and on 73% by BERTScore. Faithfulness scored 0.866, indicating that the majority of RAG responses remained consistent with the retrieved context.

Table 5. RAG evaluation results across retrieval, generation, and system-level metrics. Dashes indicate metrics not applicable to the base model.

Component	Metric	RAG	Base
Retrieval	Recall@1	0.42	—
	Recall@8	0.855	—
	MRR	0.60	—
Generation	SemScore	0.818	0.714
	SemScore Win Rate	0.815	—
	BERTScore	0.926	0.895
	BERTScore Win Rate	0.73	—
Output Quality	Faithfulness	0.866	—
	Latency (s)	3.43	2.19

Use Cases

Discussion

justify why we didn't use certain evaluation scores add latency as a limitation add negative rejection as a limitation in the evaluation process as all questions were answerable bias of using gpt-4o-mini as LLM judge Precision@k was omitted as each question has a single ground-truth chunk, making it equivalent to Recall@k scaled by 1/k and therefore uninformative.

Case Studies

Conclusion

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Table 6. Literature used to build the RAG knowledge base.

Title	Authors	Year	Topic
Book			
Encyclopedia of Marine Mammals (3rd Ed.)	Perrin et al. (eds.)	2018	Marine mammal biology
Phylogenetics & Taxonomy			
Complete Mitochondrial DNA Sequence of the Harbor Seal	Arnason & Johnsson	1992	Phylogenetics
Cytochrome b Gene of Marine Mammals: Phylogeny and Evolution	Irwin & Arnason	1994	Phylogenetics
A Molecular View of Pinniped Relationships	Arnason et al.	1995	Phylogenetics
Pinniped Phylogeny and a New Hypothesis for their Origin and Dispersal	Arnason et al.	2006	Phylogenetics
Monophyletic Origin of Sea Lions and Fur Seals (Otariidae)	Yonezawa et al.	2009	Phylogenetics
Seven New Dolphin Mitochondrial Genomes and a Time-Calibrated Phylogeny	Xiong et al.	2009	Phylogenetics
Whale Phylogeny and Rapid Radiation Events via Retroposed Elements	Chen et al.	2011	Phylogenetics
Phylogenomic Resolution of the Cetacean Tree of Life	McGowen et al.	2020	Phylogenetics
Origin and Expansion of the World's Most Widespread Pinniped	Liu et al.	2022	Phylogenetics
Ancestry Testing of "Old Tom," a Killer Whale Mutualist with Human Whalers	Reeves et al.	2023	Phylogenetics
Genomics			
De novo Assembling and Primary Analysis of the Gray Whale Genome	Moskalev et al.	2017	Genomics
Whole-Genome Sequencing of the Blue Whale and Other Rorquals	Arnason et al.	2018	Genomics
The Genome of the Steller Sea Lion	Kwan et al.	2019	Genomics
Return to the Sea, Get Huge, Beat Cancer: Humpback Whale Genome	Tollis et al.	2019	Genomics

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Title	Authors	Year	Topic
Population Genomics of Killer Whale Ecotype Evolution	Moura et al.	2014	Genomics
Genome-Culture Coevolution Promotes Rapid Divergence of Killer Whale Ecotypes	Foote et al.	2016	Genomics
Dolphin Genome Provides Evidence for Adaptive Evolution of Nervous System Genes	McGowen et al.	2012	Genomics
The Genome Sequence of the Killer Whale, <i>Orcinus orca</i>	Foote & Bunskoek	2022	Genomics
Hi-C Scaffolded Genome Assemblies of the California Sea Lion	Peart et al.	2021	Genomics
Novel <i>Megaptera novaeangliae</i> Chromosome-Level Reference Genome	Carminati et al.	2024	Genomics
Telomere-to-Telomere Genome Assembly of the Yangtze Finless Porpoise	Yin et al.	2024	Genomics
Towards Complete and Error-Free Genome Assemblies of All Vertebrate Species	Rhie et al.	2021	Genomics
Signatures of Selection on Mitonuclear Integrated Genes in Fur Seals	Vendrami et al.	2022	Genomics
Organisation and Evolution of MHC Class I Genes in Cetaceans	Day et al.	2024	Immunogenomics
Marine Adaptation			
Molecular Evolution Tracks Macroevolutionary Transitions in Cetacea	McGowen et al.	2014	Marine adaptation
Convergent Evolution of the Genomes of Marine Mammals	Foote et al.	2015	Marine adaptation
Convergent Evolution of Marine Mammals via Distinct Substitutions	Zhou et al.	2015	Marine adaptation
Molecular Footprints of Aquatic Adaptation Including Bone Mass Changes	Zhou et al.	2018	Marine adaptation
Tracing Whale Myoglobin Evolution by Resurrecting Ancient Proteins	Isogai et al.	2018	Marine adaptation
Genes Lost During the Land-to-Water Transition in Cetaceans	Huelsmann et al.	2019	Marine adaptation
Chimeric Globin Pseudogene and Aquatic Specializations in Sirenia	Signore et al.	2019	Marine adaptation

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Title	Authors	Year	Topic
Myoglobin Primary Structure Reveals Convergent Transitions to Semi-Aquatic Life	He et al.	2021	Marine adaptation
Common and Unique Strategies of Myoglobin Evolution for Deep-Sea Adaptation	Isogai et al.	2021	Marine adaptation
Comparative Genomics Provides Insights into the Aquatic Adaptations of Mammals	Yuan et al.	2021	Marine adaptation
Proteomics & Bioinformatics			
The InterPro Database for Protein Families, Domains and Functional Sites	Apweiler et al.	2001	Proteomics
Quantitative Sequence-Function Relationships in Proteins via Gene Ontology	Sangar et al.	2007	Proteomics
Functional Module Identification in Protein Interaction Networks	Wang & Qian	2014	Proteomics
Fundamentals of Protein Interaction Network Mapping	Snider et al.	2015	Proteomics
Embeddings from Deep Learning Transfer GO Annotations Beyond Homology	Littmann et al.	2020	Proteomics
Pfam: The Protein Families Database in 2021	Mistry et al.	2021	Proteomics
Proteomic Analysis of Non-Depleted Serum Proteins from Bottlenose Dolphins	Soboleski et al.	2016	Proteomics
Identification of Candidate Protein Biomarkers in California Sea Lion CSF	Ghosh et al.	2024	Proteomics
Comprehensive Expression Analysis in Bottlenose Dolphin Adipose Tissue	Suzuki et al.	2024	Transcriptomics
Genetic Evidence of Killer Whale Predation on White Sharks	Reeves et al.	2025	Ecology
Dimensionality Reduction			
Dimensionality Reduction by UMAP to Visualize Physical and Genetic Interactions	Dorrity et al.	2020	Visualization
UMAP as Dimensionality Reduction Tool for Molecular Dynamics Simulations	Trozzi et al.	2021	Visualization